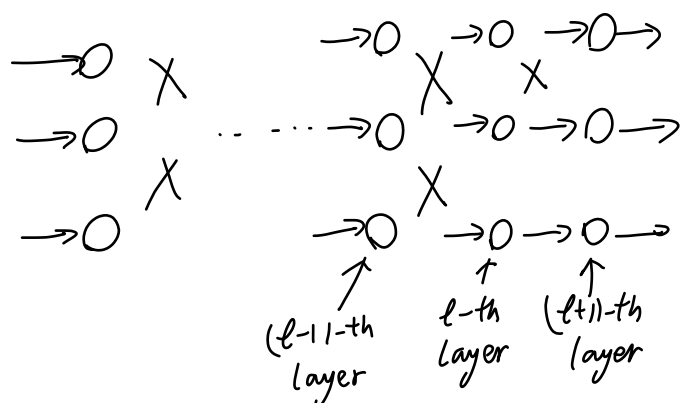


Vector Calculus: Back-propagation & Automatic Differentiation.



L : number of layer (hidden layer and output layer)

M_l : number of neurons in the l -th layer.

f_l : activation function in the l -th layer

$W^{(l)} \in \mathbb{R}^{M_l \times M_{l-1}}$: weight matrix from the $(l-1)$ -th layer to the l -th layer.

$\vec{b}^{(l)} \in \mathbb{R}^{M_l}$: $(l-1)$ -th l -th layer bias

$\vec{z}^{(l)} \in \mathbb{R}^{M_l}$: input to the l -th layer of neuron. (activation)

$\vec{a}^{(l)} \in \mathbb{R}^{M_l}$: output of the l -th layer (net activation).

Consider $\mathcal{L}(\vec{y}, \vec{\hat{y}})$ lost function to be minimized, note that

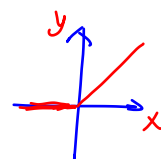
$$\vec{\hat{y}} = (f_L \circ f_{L-1} \circ \dots \circ f_1)(\vec{x}) = f_L(f_{L-1}(f_{L-2}(\dots(f_1(\vec{x}))\dots))$$

$\vec{x}$: input to the NN, $\vec{\hat{y}}$: output of the NN

f_l : activation function, $l \in [1, L]$, with $f_l(W^{(l)}\vec{a}^{(l-1)} + \vec{b}^{(l)})$ in the l -th layer.

Common activation function

- Sigmoid (S-shape curve) (Saturation on both side)
 - logistic: $\sigma(x) = \frac{1}{1+e^{-x}}$
 - Tanh: $\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$
- ReLU (rectified function)
 - original ReLU(x) = $\max(0, x)$
 - softplus(x) = $\log(1+e^x)$
 - maxout($\vec{x}$) = $\max_{k \in [1, N]} (x_k)$



Ultimate goal: obtain matrix $W^{(l)}$ and vector $\vec{b}^{(l)}$ for each layer
 $\Rightarrow$ partial derivative of $\mathcal{L}(\vec{y}, \vec{\hat{y}})$ w.r.t. $W^{(l)}$ and $\vec{b}^{(l)}$

i.e. need to compute $\frac{\partial \mathcal{L}(\vec{y}, \vec{\hat{y}})}{\partial W^{(l)}}$ and $\frac{\partial \mathcal{L}(\vec{y}, \vec{\hat{y}})}{\partial \vec{b}^{(l)}}$ for any $l \in [1, L]$.

$\Rightarrow$ backpropagation.

1. Backpropagation:

For each layer, $\vec{a}^{(l)} = f_l(\vec{z}^{(l)}) = f_l(W^{(l)}\vec{a}^{(l-1)} + \vec{b}^{(l)})$, then

$$\frac{\partial \mathcal{L}(\vec{y}, \hat{\vec{y}})}{\partial W_{ij}^{(l)}} = \frac{\partial \mathcal{L}(\vec{y}, \hat{\vec{y}})}{\partial \vec{z}^{(l)}} \cdot \frac{\partial \vec{z}^{(l)}}{\partial W_{ij}^{(l)}} \quad (\text{W.r.t. each element in } W)$$

1×1 $1 \times M_e$ $M_e \times 1$

$$\frac{1}{M_e} \frac{\partial \mathcal{L}(\vec{y}, \hat{\vec{y}})}{\partial \vec{b}^{(l)}} = \frac{1}{M_e} \frac{\partial \mathcal{L}(\vec{y}, \hat{\vec{y}})}{\partial \vec{z}^{(l)}} \cdot \frac{\partial \vec{z}^{(l)}}{\partial \vec{b}^{(l)}} \quad (\text{w.r.t. each } \vec{b})$$

$1 \times M_e$ $1 \times M_e$ $M_e \times M_e$

① computation of $\frac{\partial \vec{z}^{(l)}}{\partial W_{ij}^{(l)}} \in \mathbb{R}^{M_e \times 1}$

$$\frac{\partial \vec{z}^{(l)}}{\partial W_{ij}^{(l)}} = \begin{bmatrix} \frac{\partial z_1^{(l)}}{\partial W_{ij}^{(l)}} \\ \vdots \\ \frac{\partial z_i^{(l)}}{\partial W_{ij}^{(l)}} \\ \vdots \\ \frac{\partial z_{M_e}^{(l)}}{\partial W_{ij}^{(l)}} \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ \frac{\partial (W_{ij}^{(l)} \vec{a}^{(l-1)} + \vec{b}^{(l)})}{\partial W_{ij}^{(l)}} \\ \vdots \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ a_j^{(l-1)} \\ \vdots \\ 0 \end{bmatrix} = \prod_i (a_j^{(l-1)})$$

$\uparrow$
 the i -th element is $a_j^{(l-1)}$, 0 otherwise.

② computation of $\frac{\partial \vec{z}^{(l)}}{\partial \vec{b}^{(l)}} \in \mathbb{R}^{M_e \times M_e}$.

since $\vec{z}^{(l)} = W^{(l)} \cdot \vec{a}^{(l-1)} + \vec{b}^{(l)} \Rightarrow \frac{\partial \vec{z}^{(l)}}{\partial \vec{b}^{(l)}} = I_{M_e}$ ✓
 with I_{M_e} being an $M_e \times M_e$ identity matrix.

③ computation of $\frac{\partial \mathcal{L}(\vec{y}, \hat{\vec{y}})}{\partial \vec{z}^{(l)}}$

Note: representing the influence of the l -th layer of neurons on the final loss (we call it error term of the l -th layer)

denoted by $\vec{\delta}^{(l)} := \frac{\partial \mathcal{L}(\vec{y}, \hat{\vec{y}})}{\partial \vec{z}^{(l)}} \in \mathbb{R}^{1 \times M_e}$.

$$\vec{\delta}^{(l)} = \frac{\partial L(\vec{y}, \hat{\vec{y}})}{\partial \vec{z}^{(l+1)}} \cdot \frac{\partial \vec{z}^{(l+1)}}{\partial \vec{a}^{(l)}} \cdot \frac{\partial \vec{a}^{(l)}}{\partial \vec{z}^{(l)}}$$

Dimensions: $1 \times M_l$ (green), $1 \times M_{l+1}$ (green), $M_{l+1} \times M_l$ (red), $M_{l+1} \times M_l$ (red), $M_l \times M_l$ (red). ✓

Since $\vec{z}^{(l+1)} = W^{(l+1)} \vec{a}^{(l)} + b^{(l+1)}$

$$\Rightarrow \frac{\partial \vec{z}^{(l+1)}}{\partial \vec{a}^{(l)}} = W^{(l+1)}$$

additionally, since $\vec{a}^{(l)} = f_l(\vec{z}^{(l)})$

$$\Rightarrow \frac{\partial \vec{a}^{(l)}}{\partial \vec{z}^{(l)}} = \frac{\partial f_l(\vec{z}^{(l)})}{\partial \vec{z}^{(l)}} = \text{diag}(f'_{l,1}(\vec{z}^{(l)}) \dots f'_{l,M_l}(\vec{z}^{(l)}))$$

$$\Rightarrow \vec{\delta}^{(l)} = \frac{\partial L(\vec{y}, \hat{\vec{y}})}{\partial \vec{z}^{(l+1)}} \cdot \frac{\partial \vec{z}^{(l+1)}}{\partial \vec{a}^{(l)}} \cdot \frac{\partial \vec{a}^{(l)}}{\partial \vec{z}^{(l)}}$$

$$= \vec{\delta}^{(l+1)} \cdot W^{(l+1)} \cdot \text{diag}(\dots)$$

$\underline{f_l}$: if the activation function is the same in the l -th layer

$$\vec{\delta}^{(l)} = \vec{\delta}^{(l+1)} \cdot W^{(l+1)} \cdot \text{diag}(f'_l(\vec{z}^{(l)}) \dots f'_l(\vec{z}^{(l)}))$$

$$= f'_l(\vec{z}^{(l)}) \odot (\vec{\delta}^{(l+1)} \cdot W^{(l+1)})$$

$\odot$: Hadamard product:
elementwise product

Def: $A, B \in \mathbb{R}^{m \times n}$ $[A \odot B]_{mn} = a_{mn} \cdot b_{mn}$

a_{mn} : (m, n) -th element in A

b_{mn} : (m, n) -th element in B

$c \in \mathbb{R}$, $A \in \mathbb{R}^{m \times n}$, $[c \odot A]_{mn} = c a_{mn}$

$$\Rightarrow \frac{\partial \mathcal{L}(\vec{y}, \vec{\hat{y}})}{\partial w_{ij}^{(l)}} = \underbrace{\begin{bmatrix} f_1^{(l)} & \dots & f_{M_L}^{(l)} \end{bmatrix}}_{\vec{f}^{(l)}} \begin{bmatrix} 0 \\ \vdots \\ a_j^{(l-1)} \\ \vdots \\ 0 \end{bmatrix} = \underline{f_i^{(l)} \cdot a_j^{(l-1)}}$$

Remark: $\vec{a} \in \mathbb{R}^M$, $\vec{b} \in \mathbb{R}^N$
with $\vec{a} = [a_1, \dots, a_M]^T$
 $\vec{b} = [b_1, \dots, b_N]^T$

the (i, j) -th element of the outer product between $\vec{f}^{(l)}$ and $\vec{a}^{(l-1)}$

$$\vec{a} \otimes \vec{b} = \begin{bmatrix} a_1 b_1 & a_1 b_2 & \dots & a_1 b_N \\ a_2 b_1 & a_2 b_2 & \dots & a_2 b_N \\ \vdots & \vdots & \ddots & \vdots \\ a_M b_1 & a_M b_2 & \dots & a_M b_N \end{bmatrix} = \vec{a} \vec{b}^T, \text{ with } [\vec{a} \otimes \vec{b}]_{mn} = a_m b_n.$$

$$\Rightarrow \left[\frac{\partial \mathcal{L}(\vec{y}, \vec{\hat{y}})}{\partial w^{(l)}} \right]_{ij} = [\vec{f}^{(l)} \otimes \vec{a}^{(l-1)}]_{ij} \Rightarrow \frac{\partial \mathcal{L}(\vec{y}, \vec{\hat{y}})}{\partial w^{(l)}} = \vec{f}^{(l)} \cdot \vec{a}^{(l-1)T}$$

Moreover

$$\frac{\partial \mathcal{L}(\vec{y}, \vec{\hat{y}})}{\partial \vec{b}^{(l)}} = \frac{\partial \mathcal{L}(\vec{y}, \vec{\hat{y}})}{\partial \vec{z}^{(l)}} \cdot \frac{\partial \vec{z}^{(l)}}{\partial \vec{b}^{(l)}} = \begin{bmatrix} f_1^{(l)} & \dots & f_{M_L}^{(l)} \end{bmatrix} \cdot \underbrace{\begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}}_{M_L \times M_L} = \vec{f}^{(l)} \in \mathbb{R}^{1 \times M_L}$$

Summarize: procedure for training with backpropagation

(1) select data from the training set $(\vec{x}^{(n)}, \vec{y}^{(n)})$

(2) Feed forward computation of each activation $\vec{z}^{(l)}$ and the net activation $\vec{a}^{(l)}$

(3) compute error $\vec{\delta}^{(l)}$, $l \in [1, L]$, on each layer via backpropagation

$$\vec{\delta}^{(l)} = \frac{\partial L(\vec{y}, \hat{\vec{y}})}{\partial \vec{z}^{(l)}} = \vec{\delta}^{(l+1)} \cdot \frac{\partial L(\vec{y}, \hat{\vec{y}})}{\partial \vec{a}^{(l+1)}} \cdot \frac{\partial \vec{a}^{(l+1)}}{\partial \vec{z}^{(l)}}$$

$$= \vec{\delta}^{(l+1)} \cdot W^{(l+1)} \cdot \text{diag}(f'_{l_1}(\vec{z}^{(l+1)}), \dots, f'_{l_{n_{l+1}}}(\vec{z}^{(l+1)}))$$

for $l \in [1, L-1]$ recursively.

(4) compute gradient of $W^{(l)}$ and $\vec{b}^{(l)}$ for all layer l

$$\frac{\partial L(\vec{y}, \hat{\vec{y}})}{\partial W^{(l)}} = \vec{\delta}^{(l)} (\vec{a}^{(l-1)})^T, \quad \frac{\partial L(\vec{y}, \hat{\vec{y}})}{\partial \vec{b}^{(l)}} = \vec{\delta}^{(l)}$$

(5) update W and $\vec{b}$ based on the gradient (to be discussed)

2. Vanishing (exploding) gradient.

(1) consider the same activation function in each layer.

$$\vec{\delta}^{(l)} = f'_L(\vec{z}^{(L)}) \odot (\vec{\delta}^{(L+1)} \cdot W^{(L+1)})$$

$$\vec{\delta}^{(l+1)} = f'_L(\vec{z}^{(l+1)}) \cdot \vec{\delta}^{(l+2)} \cdot W^{(l+2)}$$

$$\vec{\delta}^{(l)} = f'_L(\vec{z}^{(l)}) \cdot f'_L(\vec{z}^{(l+1)}) \cdot \vec{\delta}^{(l+1)} \cdot W^{(l+2)}$$

(1) if $f'_L(\vec{z}^{(l)}) < 1 \Rightarrow L \uparrow, \vec{\delta}^{(l)} \rightarrow 0$ (consider $0.9^{1000} = 4.3 \times 10^{-5}$)
 $\Rightarrow$ gradient tends to 0 (vanishing gradient)

(2) if $f'_L(\vec{z}^{(l)}) > 1 \Rightarrow L \uparrow, \vec{\delta}^{(l)} \rightarrow \infty$ (grow exponentially)
 gradient $\rightarrow \infty$ (exploding)

solution { Gradient clipping
 Batch normalization
 Proper Initialization of parameters

3. Automatic differentiation:
consider the function.

$$f(x) = \sqrt{x^2 + e^{x^2}} + \cos(x^2 + e^{x^2}), \text{ what is the derivative}$$

$$\frac{df}{dx} \text{ evaluated at } x=1$$

Typical solution:

$$\frac{df}{dx} = \frac{2x + 2e^{x^2}}{2\sqrt{x^2 + e^{x^2}}} - \sin(x^2 + e^{x^2})(2x + 2xe^{x^2}) \Big|_{x=1}$$

Expensive and even impossible (consider nesting functions)

$\Rightarrow$ automatic differentiation over computational graph
consider

$$a = x^2$$

$$e^{x^2} = b = \exp(a)$$

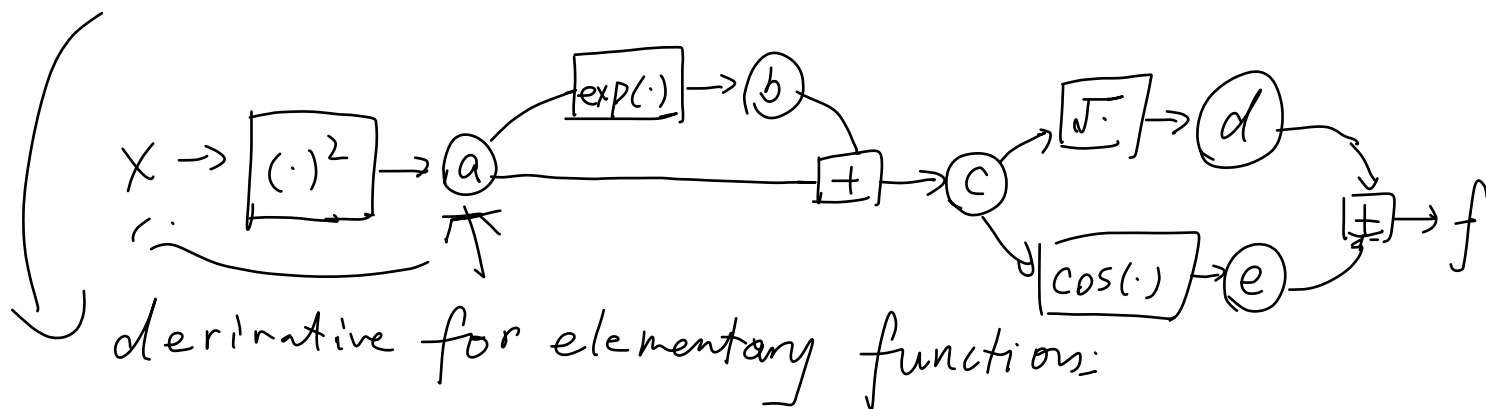
$$c = a + b$$

$$d = \sqrt{c}$$

$$e = \cos(c)$$

$$f = d + e$$

$\Rightarrow$
computation
graph.



$$\frac{\partial a}{\partial x} = 2x$$

$$\frac{\partial b}{\partial a} = \exp(a)$$

$$\frac{\partial c}{\partial a} = 1 = \frac{\partial c}{\partial b}$$

d

$$\frac{\partial d}{\partial c} = \frac{1}{2\sqrt{c}} \quad \frac{\partial e}{\partial c} = -\sin(c)$$

$$\frac{\partial f}{\partial d} = 1 = \frac{\partial f}{\partial e}$$

$$\begin{aligned} \frac{\partial f}{\partial c} &= \frac{\partial f}{\partial d} \cdot \frac{\partial d}{\partial c} + \frac{\partial f}{\partial e} \cdot \frac{\partial e}{\partial c} = 1 \cdot \frac{1}{2\sqrt{c}} + 1 \cdot (-\sin(c)) \\ &= \frac{1}{2\sqrt{c}} - \sin(c) \end{aligned}$$

$$\frac{\partial f}{\partial b} = \frac{\partial f}{\partial c} \cdot \frac{\partial c}{\partial b} = \frac{\partial f}{\partial c}$$

$$\frac{\partial f}{\partial a} = \frac{\partial f}{\partial b} \cdot \frac{\partial b}{\partial a} + \frac{\partial f}{\partial c} \cdot \frac{\partial c}{\partial a} = \frac{\partial f}{\partial b} \cdot \exp(a) + \frac{\partial f}{\partial c} \cdot 1$$

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial a} \cdot \frac{\partial a}{\partial x} = \frac{\partial f}{\partial a} \cdot 2x$$

$$= \frac{\partial f}{\partial c} (\exp(a) + 1) \cdot 2x$$

$$= \left(\frac{1}{2\sqrt{c}} - \sin(c) \right) (\exp(a) + 1) \cdot 2x$$

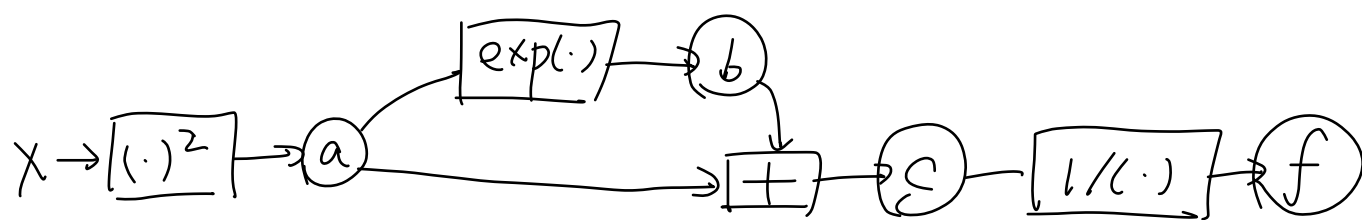
$\Downarrow$ one can verify $\Rightarrow$ easy to compute.

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial d} \cdot \frac{\partial d}{\partial c} \cdot \frac{\partial c}{\partial b} \cdot \frac{\partial b}{\partial a} \cdot \frac{\partial a}{\partial x} + \frac{\partial f}{\partial e} \cdot \frac{\partial d}{\partial c} \cdot \frac{\partial c}{\partial a} \cdot \frac{\partial a}{\partial x} +$$

$$\frac{\partial f}{\partial e} \cdot \frac{\partial e}{\partial c} \cdot \frac{\partial c}{\partial b} \cdot \frac{\partial b}{\partial a} \cdot \frac{\partial a}{\partial x} + \frac{\partial f}{\partial e} \cdot \frac{\partial e}{\partial c} \cdot \frac{\partial c}{\partial a} \cdot \frac{\partial a}{\partial x}$$

Exercise

$$f = \frac{1}{\exp(x^2) + x^2} \quad \text{compute } \frac{\partial f}{\partial x} \Big|_{x=1}$$



$$\Rightarrow \frac{\partial f}{\partial x} = \frac{\partial f}{\partial c} \cdot \frac{\partial c}{\partial a} \cdot \frac{\partial a}{\partial x} + \frac{\partial f}{\partial c} \cdot \frac{\partial c}{\partial b} \cdot \frac{\partial b}{\partial a} \cdot \frac{\partial a}{\partial x}$$

$$= -\frac{1}{c^2} \cdot 1 \cdot 2x + \left(-\frac{1}{c^2}\right) \cdot 1 \cdot \exp(a) \cdot 2x$$

$$= -\frac{2x}{c^2} (1 + \exp(a)), \text{ 代 \lambda 进 } \frac{\partial f}{\partial x}$$

note:

$$a = x^2 \quad \frac{\partial a}{\partial x} = 2x$$

$$b = \exp(a) \quad \frac{\partial b}{\partial a} = e^a$$

$$c = a + b$$

$$\frac{\partial c}{\partial a} = 1 = \frac{\partial c}{\partial b}$$

$$\frac{\partial f}{\partial c} = -\frac{1}{c^2}$$